A/C HEATER CONTROL (Continued)

DIAGNOSIS AND TESTING

VACUUM CONTROL SYSTEM

Vacuum control is used to operate the mode-air doors within the HVAC housing. Testing the operation of the heating-only and heating-A/C system mode control will determine if the vacuum and mechanical controls are functioning. However, it is possible that a vacuum control system that operates correctly at engine idle (high engine vacuum) may not function properly at high engine speeds or loads (low engine vacuum). This can be caused by leaks within the vacuum system, or a faulty vacuum check valve.

A vacuum system test will help to identify the source of poor vacuum system performance or vacuum system leaks. Before starting this test, stop the engine and make certain that the problem is not a disconnected vacuum supply line at the engine intake manifold vacuum tap or at the vacuum reservoir.

Use an adjustable vacuum test set (Special Tool C-3707-B) and a suitable vacuum pump to test the HVAC vacuum control system. With a finger placed over the end of the vacuum test hose probe (Fig. 12), adjust the bleed valve on the test set gauge to obtain a vacuum of exactly 27 kPa (8 in. Hg.). Release and block the end of the probe several times to verify that the vacuum reading returns to the exact 27 kPa (8 in. Hg.) setting. Otherwise, a false reading will be obtained during testing.

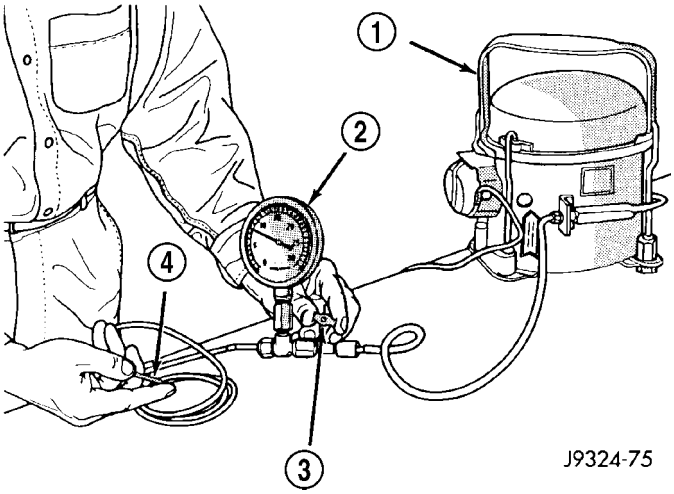

Fig. 12 Adjust Vacuum Test Bleed Valve

1 - VACUUM PUMP TOOL C-4289
2 - VACUUM TEST SET C-3707
3 - BLEED VALVE
4 - PROBE

VACUUM CHECK VALVE

(1) Remove the vacuum check valve. The valve is located in the (black) vacuum supply tube at the intake manifold vacuum tap.

(2) Connect the test set vacuum supply hose to the A/C-heater control side of the vacuum check valve. When connected to this side of the check valve, no vacuum should pass and the test set gauge should return to the 27 kPa (8 in. Hg.) setting. If OK, go to Step 3. If not OK, replace the faulty vacuum check valve.

(3) Connect the test set vacuum supply hose to the engine vacuum side of the vacuum check valve. When connected to this side of the check valve, vacuum should flow through the check valve without restriction. If not OK, replace the faulty vacuum check valve.

A/C HEATER CONTROLS

(1) Connect the test set vacuum probe to the heater-A/C vacuum supply (black) tube in the engine compartment. Position the test set gauge so that it can be viewed from the passenger compartment.

(2) Place the A/C-heater mode control in each mode position, one position at a time, and pause after each selection. The test set gauge should return to the 27 kPa (8 in. Hg.) setting shortly after each selection is made. If not OK, a component or vacuum line in the vacuum circuit of the selected mode has a leak. See the procedure in Locating Vacuum Leaks.

LOCATING VACUUM LEAKS

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the battery negative (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. This is the only sure way to disable the airbag system. Failure to take the proper precautions could result in accidental airbag deployment and possible personal injury or death.

CAUTION: Do not use lubricant on the switch ports or in the holes of the connector, as lubricant will ruin the vacuum valve in the switch. A drop of clean water in the connector holes will help the connector slide onto the switch ports.